Evidence Preservation

---

1. Introduction to Evidence Preservation

What is Evidence Preservation?

Evidence preservation is the process of collecting, securing, documenting, and maintaining digital evidence in a manner that ensures its integrity and admissibility during investigations.

In cybersecurity incident response, evidence preservation helps investigators identify the cause of an incident, determine the extent of compromise, and support legal or compliance requirements.

Objectives of Evidence Preservation

- Maintain evidence integrity
- Prevent evidence tampering
- Support forensic investigations
- Enable incident reconstruction
- Ensure legal admissibility
- Maintain chain of custody

---

2. Types of Digital Evidence

Volatile Evidence

Volatile evidence is lost when a system is powered off.

Examples:

- RAM contents
- Running processes
- Active network connections
- Logged-in users
- Open files

Importance

Volatile data should be collected first because it disappears after shutdown.

Non-Volatile Evidence

Non-volatile evidence remains stored even after power loss.

Examples:

- Hard drives
- SSDs
- Log files
- Configuration files
- Registry entries

Importance

Provides long-term evidence regarding attacker activity.

---

3. Evidence Collection Procedure

Step 1: Identify Evidence

Determine systems involved in the incident.

Example:

- Server-X
- Employee Workstation
- Firewall Logs

Step 2: Preserve the Scene

Prevent modifications to the affected system.

Actions:

- Disconnect from network
- Restrict access
- Avoid rebooting

Step 3: Collect Volatile Data

Examples:

Active Connections

netstat -ano

Running Processes

tasklist

Logged-In Users

whoami

Step 4: Collect Non-Volatile Data

Examples:

- Disk Images
- System Logs
- Application Logs
- Security Logs

Step 5: Generate Hash Values

Hashes verify integrity.

Example:

SHA256:
4f7f6f9a8c1e6d9f8a4e5b7c9d2f1a6e

Step 6: Document Collection

Record:

- Collector name
- Date and time
- Evidence source
- Hash value

---

4. Chain of Custody

Definition

Chain of Custody is a formal record showing how evidence was collected, handled, transferred, stored, and analyzed.

It ensures evidence integrity throughout the investigation.

Importance

- Prevents tampering
- Maintains accountability
- Supports legal proceedings
- Ensures transparency

Chain of Custody Form

Evidence ID	Description	Collected By	Date & Time	Hash Value
EV-001	Memory Dump	SOC Analyst	29-May-2026 10:00 AM	SHA256-ABC123
EV-002	Security Logs	SOC Analyst	29-May-2026 10:20 AM	SHA256-XYZ456
EV-003	Disk Image	SOC Analyst	29-May-2026 11:00 AM	SHA256-DEF789

---

5. Memory Dump Collection

Objective

Capture system memory for forensic analysis.

Tool Used

Velociraptor

Collection Command

SELECT * FROM Artifact.Windows.Memory.Acquisition

Collected Information

- Running processes
- Injected code
- Active malware
- Network connections

Collection Record

Item	Value
Evidence ID	EV-001
Evidence Type	Memory Dump
Collected By	SOC Analyst
Date	29-May-2026

Item	Value
Hash	SHA256-ABC123

---

6. Network Connection Collection

Objective

Identify suspicious communication channels.

Tool Used

Velociraptor

Query

SELECT * FROM netstat()

Sample Results

Protocol	Local Address	Foreign Address	State
TCP	192.168.1.10:443	203.0.113.25:8080	ESTABLISHED
TCP	192.168.1.10:22	192.168.1.100:50200	ESTABLISHED

Analysis

Connection to unknown external IP identified and flagged for investigation.

---

7. Evidence Inventory

Evidence ID	Evidence Type	Description	Storage Location
EV-001	Memory Dump	Server-X Memory	Evidence Vault
EV-002	Firewall Logs	Firewall Activity Logs	Evidence Vault
EV-003	Disk Image	Workstation Disk Copy	Evidence Vault
EV-004	Network Logs	Traffic Analysis Logs	Evidence Vault

---

8. Evidence Preservation Checklist

Identification

- Incident confirmed
- Evidence sources identified

Collection

- Volatile data collected
- Memory dump captured
- Logs exported

Integrity Verification

- Hash values generated
- Evidence validated

Documentation

- Collection details recorded
- Chain of custody completed

Storage

- Evidence secured
- Access restricted

---

9. Sample Forensic Report

Incident Summary

A suspicious outbound network connection was detected from Server-X to an unknown external IP address. Evidence collection procedures were initiated to preserve critical forensic artifacts.

Evidence Collected

1. Memory Dump
2. Security Logs
3. Firewall Logs
4. Network Connections

5. Disk Image

Findings

- Unauthorized outbound communication observed.
- Suspicious process execution identified.
- Evidence preserved successfully.

Recommendations

1. Block malicious IP address.
2. Conduct full malware scan.
3. Review firewall rules.
4. Continue monitoring affected systems.

---

10. Conclusion

Evidence preservation is a critical component of incident response and digital forensics. Proper collection, hashing, documentation, and chain-of-custody procedures ensure that evidence remains reliable and legally defensible. By following established forensic practices, organizations can effectively investigate incidents while maintaining the integrity of digital evidence.